AP CALCULUS AB

Unit 1 • Lesson 1-1 • Notes (filled in)

Lesson 1-1 • Limits to a Point on a Graph

Notes (filled in)

Learning goals

- Read and write limit notation, including one-sided limits.
- Find limits from a graph and decide whether a limit exists.
- Distinguish the limit of f at c from the value $f(c)$.

Vocabulary. limit, left-hand limit, right-hand limit, open circle, closed (filled) dot, does not exist (DNE), unbounded, oscillating.

1. What a limit means

Key idea

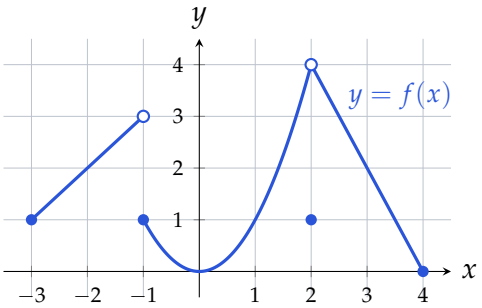
$\lim_{x\to c} f(x) = L$ means: as x gets closer and closer to __c__ (from both sides, with $x \neq c$), the values $f(x)$ get closer and closer to __L__. The value of f *at* c __does not affect__ the limit; $f(c)$ may even be undefined.

2. One-sided limits

$\lim_{x\to c^-} f(x)$ is the __left-hand__ limit (uses $x < c$); $\lim_{x\to c^+} f(x)$ is the __right-hand__ limit (uses $x > c$).

When does a limit exist?

$\lim_{x\to c} f(x) = L$ exists if and only if $\lim_{x\to c^-} f(x)$ and $\lim_{x\to c^+} f(x)$ both exist and are __equal to L__.



Example 1 • Reading the graph of f

Complete the table.

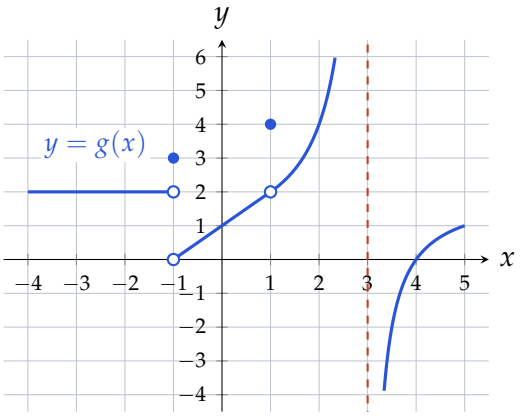
c	$\lim_{x\to c^-} f(x)$	$\lim_{x\to c^+} f(x)$	$\lim_{x\to c} f(x)$	$f(c)$
-1	3	1	DNE	1
0	0	0	0	0
2	4	4	4	1
3	2	2	2	2
-3	none	1	one-sided only	1

3. Three ways a limit fails to exist

Key idea

- **Jump**: the one-sided limits are different.
- **Unbounded behavior**: $f(x)$ increases or decreases without bound (vertical asymptote).
- **Oscillation**: $f(x)$ keeps oscillating between values, like $\sin\frac{1}{x}$ near 0.

Example 2 • A graph with an asymptote



Find $\lim_{x\to -1} g(x)$, $\lim_{x\to 1} g(x)$, $\lim_{x\to 3^-} g(x)$, $\lim_{x\to 3^+} g(x)$, and $\lim_{x\to 0} g\big(g(x)\big)$.

At -1: left 2, right 0: DNE. At 1: both sides approach 2: the limit is 2 (although $g(1) = 4$).
$\lim_{x\to 3^-} g(x) = +\infty$, $\lim_{x\to 3^+} g(x) = -\infty$.
As $x \to 0$, $g(x) = x + 1 \to 1$ (never equal to 1), so $g(g(x)) \to \lim_{u\to 1} g(u) = 2$.

Common mistakes

- Reading the filled dot as the limit.
- Writing "$\lim f(x) =$ DNE". Write "the limit does not exist" and give the reason.

Check your understanding

1. If $\lim_{x\to 4^-} h = 5$, $\lim_{x\to 4^+} h = 5$ and $h(4) = 0$, then $\lim_{x\to 4} h(x) =$ 5.
2. If $\lim_{x\to 1^-} k = 2$ and $\lim_{x\to 1^+} k = -2$, then $\lim_{x\to 1} k(x)$ does not exist.